

Aerosense

AI-Driven Spatiotemporal Air Quality Forecasting Engine for Delhi-NCR: Ablation Study of Recurrent and Convolutional Architectures

Yashaswi Garg
Student, Bennett University, 2024–2028

Abstract—Delhi and the National Capital Region (NCR) of India consistently rank among the most severely polluted urban agglomerations on Earth, with fine particulate matter and gaseous pollutants posing acute public-health risks across hundreds of millions of residents. Accurate, spatially resolved multi-day air quality forecasting is therefore a critical societal need, yet existing approaches either rely on coarse-resolution numerical chemical-transport models or on point-sensor data that cannot capture the full spatial extent of pollution plumes. In this work we present a deep-learning forecasting engine that ingests an eight-year (2018–2025) archive of satellite-derived pollutant observations from the TROPOMI instrument aboard Sentinel-5P, regridded to a 141×231 pixel spatial lattice over Delhi-NCR, and ingests a 7-day historical sequence to predict the complete multi-channel concentration field for the subsequent 24 hours on a pixel-wise basis. The six modelled channels are nitrogen dioxide (NO_2), carbon monoxide (CO), Aerosol Optical Depth (AOD), UV Aerosol Index (UVAI), and the ERA5 horizontal wind components (U, V), providing a physically complete representation of both emission-source signatures and advective transport. We conduct a systematic ablation study evaluating seven deep learning architectures spanning recurrent convolutional and purely spatial paradigms: (i) a Standard ConvLSTM baseline, (ii) an Advanced ConvLSTM augmented with lagged wind features and a physics-weighted loss, (iii) a custom Deep Convolutional Gated Recurrent Unit (Conv-GRU) with a manually unrolled bi-gate loop, and (iv–vii) four spatial convolutional architectures adapted for regression—AlexNet, ResNet-50, ResNet-101, and U-Net. On a 20% hold-out test partition spanning approximately 500 days, the Conv-GRU achieves state-of-the-art performance of $R^2=0.8305$, $\text{MAE}=0.001381$, $\text{RMSE}=0.001974$, and $\text{MAPE}=3.37\%$, outperforming all six competing architectures. A notable secondary finding is that AlexNet ($R^2=0.7200$) significantly outperforms ResNet-50 ($R^2=0.6879$) and ResNet-101 ($R^2=0.6323$) despite its lower representational capacity, a result we attribute to its large initial receptive field acting as a macro-level spatial filter and to severe over-parameterisation in the ResNet variants under our data regime. We argue that across all architectures tested, structural parsimony consistently outperforms expressive capacity when training on noisy satellite retrievals with approximately 2,000 samples.

Index Terms—Air quality forecasting, spatiotemporal deep learning, convolutional GRU, ConvLSTM, AlexNet, ResNet, U-Net, TROPOMI, Sentinel-5P, Delhi-NCR, advection, ablation study, satellite remote sensing.



1 INTRODUCTION

THE Indo-Gangetic Plain, on which Delhi and its satellite cities sit, experiences some of the most extreme episodes of atmospheric pollution recorded anywhere in the world. The Indian capital routinely exceeds the World Health Organization’s 24-hour guideline for $\text{PM}_{2.5}$ by factors of ten to twenty during the winter months, driven by a convergence of factors that span multiple spatial and temporal scales.

Emission Sources and Seasonality

At the micro-scale, vehicular exhaust from the estimated twelve million registered vehicles in Delhi contributes continuous baseline loadings of nitrogen oxides (NO_x), CO, and fine carbonaceous aerosols throughout the year. Construction dust and industrial emissions from the NCR peripheral belt add mineral and metallic aerosol components. Superimposed on this chronic background are acute episodic contributions. The most significant is post-harvest

agricultural residue burning in the states of Punjab and Haryana, which typically peaks between mid-October and mid-November and generates enormous quantities of BC (black carbon), organic carbon, and gaseous co-pollutants whose plumes are subsequently advected into Delhi by the prevailing north-westerly wind regime. Diwali fireworks provide a further acute nocturnal source. The combination of these source categories, occurring under radiation-fog conditions that suppress boundary-layer mixing heights to as little as 100–200 m, routinely produces 24-hour average $\text{PM}_{2.5}$ concentrations exceeding $500 \mu\text{g m}^{-3}$.

The Physics of Advective Transport

Understanding why spatial deep learning is appropriate for this problem requires appreciating the dominant atmospheric dynamics at play. Under synoptic-scale westerly or north-westerly flow, pollution plumes from out-of-domain agricultural burning are advected coherently across the Indo-Gangetic Plain at horizontal velocities of $5\text{--}15 \text{ m s}^{-1}$. This

advection process—the bulk transport of a scalar quantity ϕ by the vector wind field $\mathbf{u}$ —is governed by the advection equation:

$$\frac{\partial \phi}{\partial t} + \mathbf{u} \cdot \nabla \phi = S - L, \quad (1)$$

where S and L denote source and loss terms respectively. A model that encodes spatial context (the gradient $\nabla \phi$) alongside temporal history (via recurrent hidden states) is therefore physically motivated.

Gaps in Existing Approaches

Numerical chemical-transport models (CTMs) such as WRF-Chem and CMAQ can resolve these dynamics but require high-resolution meteorological driving fields, chemical boundary conditions, and emission inventories that are difficult to compile with sufficient accuracy and temporal currency for operational use. Ground-based monitoring networks (e.g., the Central Pollution Control Board’s CAAQMS network) provide high-temporal-resolution point observations but are sparse over the NCR domain and entirely absent over agricultural source regions.

Spatiotemporal deep learning offers an attractive middle path: given sufficient observational data (here, eight years of continuous satellite retrievals), a neural network can learn the effective dynamical operator implicit in Eq. (1) directly from data, without requiring explicit specification of emissions or chemistry [10]. However, the choice of architecture strongly conditions whether the learned operator reflects physical dynamics or merely memorises dataset-specific artefacts. Prior work has demonstrated the utility of convolutional recurrent models for precipitation nowcasting [3] and environmental forecasting [16], but a systematic comparison of recurrent versus purely spatial architectures under noisy satellite conditions has not been reported for the Delhi-NCR domain. This ablation study is designed to isolate that architectural signal across seven candidate architectures.

2 BACKGROUND

2.1 Urban Air Pollution Dynamics

The atmospheric boundary layer over Delhi exhibits strong diurnal variability. During daytime, solar heating drives convective mixing that dilutes surface concentrations and lifts pollutants aloft. At sunset the boundary layer collapses under radiative cooling, trapping emitted and advected pollutants within a shallow nocturnal layer. This diurnal oscillation, superimposed on synoptic-scale advection and modulated by mesoscale circuitous flows around the Aravalli ridge, produces a highly non-stationary, spatially heterogeneous concentration field that resists prediction by simple statistical methods.

Chemically, the dominant species exhibit distinct spatial signatures. NO_2 has a short atmospheric lifetime ($\sim$ hours) and therefore traces fresh emission sources with high spatial fidelity—roads, power plants, and industrial zones appear as hotspots. CO has a longer lifetime ($\sim$ months) and is therefore more uniformly mixed, serving as a tracer of regional-scale combustion (including agricultural fires). AOD integrates over the full aerosol column and is sensitive to both primary

$\text{PM}_{2.5}$ and secondary aerosol formation. UAVI detects UV-absorbing aerosols, particularly black and brown carbon from biomass burning, and is thus especially diagnostic of stubble-burning plumes. Together, these six channels provide complementary physical information that no single channel alone could supply.

2.2 Recurrent Neural Networks for Sequential Spatial Data

Standard Recurrent Neural Networks (RNNs) process a sequence $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T\}$ by maintaining a hidden state $\mathbf{h}_t$ that evolves according to:

$$\mathbf{h}_t = \sigma(\mathbf{W}_h \mathbf{h}_{t-1} + \mathbf{W}_x \mathbf{x}_t + \mathbf{b}), \quad (2)$$

where σ is a non-linear activation and $\mathbf{W}_h, \mathbf{W}_x$ are weight matrices. Plain RNNs suffer from vanishing/exploding gradients over long sequences, motivating gated architectures [14].

2.2.1 Long Short-Term Memory (LSTM)

The LSTM [9] augments the hidden state with a *cell state* $\mathbf{c}_t$ and three multiplicative gates:

$$\mathbf{i}_t = \sigma(\mathbf{W}_{xi} \mathbf{x}_t + \mathbf{W}_{hi} \mathbf{h}_{t-1} + \mathbf{b}_i) \quad (3)$$

$$\mathbf{f}_t = \sigma(\mathbf{W}_{xf} \mathbf{x}_t + \mathbf{W}_{hf} \mathbf{h}_{t-1} + \mathbf{b}_f) \quad (4)$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_{xo} \mathbf{x}_t + \mathbf{W}_{ho} \mathbf{h}_{t-1} + \mathbf{b}_o) \quad (5)$$

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_{xc} \mathbf{x}_t + \mathbf{W}_{hc} \mathbf{h}_{t-1} + \mathbf{b}_c) \quad (6)$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \quad (7)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t). \quad (8)$$

The three-gate structure endows the LSTM with substantial representational capacity, but also substantially increases parameter count, particularly when the inputs $\mathbf{x}_t$ are high-dimensional spatial tensors.

2.2.2 Gated Recurrent Unit (GRU)

The GRU [5] consolidates the forget and input gates into a single *update gate* and introduces a *reset gate*, requiring only two gate computations:

$$\mathbf{Z}_t = \sigma(\mathbf{W}_{xz} \mathbf{x}_t + \mathbf{W}_{hz} \mathbf{h}_{t-1} + \mathbf{b}_z) \quad (9)$$

$$\mathbf{R}_t = \sigma(\mathbf{W}_{xr} \mathbf{x}_t + \mathbf{W}_{hr} \mathbf{h}_{t-1} + \mathbf{b}_r) \quad (10)$$

$$\tilde{\mathbf{H}}_t = \tanh(\mathbf{W}_{xh} \mathbf{x}_t + \mathbf{W}_{hh} (\mathbf{R}_t \odot \mathbf{h}_{t-1}) + \mathbf{b}_h) \quad (11)$$

$$\mathbf{H}_t = (1 - \mathbf{Z}_t) \odot \mathbf{h}_{t-1} + \mathbf{Z}_t \odot \tilde{\mathbf{H}}_t. \quad (12)$$

This structural parsimony is not merely aesthetic: fewer parameters reduce overfitting risk—a property we exploit in the convolutional generalisation described in Section 4.

2.2.3 Convolutional Extension

Shi *et al.* [3] proposed replacing the matrix-vector multiplications in LSTM gates with 2-D convolutions, yielding ConvLSTM. Each gate kernel simultaneously processes the spatial neighbourhood and the channel axis, producing a spatially coherent gating signal that respects the topology of the underlying concentration field. The analogous Conv-GRU

replaces the matrix products in Eqs. (9)–(12) with convolution operators $*$:

$$\mathbf{Z}_t = \sigma(\mathbf{W}_{xz} * \mathbf{X}_t + \mathbf{W}_{hz} * \mathbf{H}_{t-1} + \mathbf{b}_z), \quad (13)$$

$$\mathbf{R}_t = \sigma(\mathbf{W}_{xr} * \mathbf{X}_t + \mathbf{W}_{hr} * \mathbf{H}_{t-1} + \mathbf{b}_r), \quad (14)$$

$$\tilde{\mathbf{H}}_t = \tanh(\mathbf{W}_{xh} * \mathbf{X}_t + \mathbf{W}_{hh} * (\mathbf{R}_t \odot \mathbf{H}_{t-1}) + \mathbf{b}_h), \quad (15)$$

$$\mathbf{H}_t = (1 - \mathbf{Z}_t) \odot \mathbf{H}_{t-1} + \mathbf{Z}_t \odot \tilde{\mathbf{H}}_t. \quad (16)$$

Here all quantities are 3-D tensors of shape (H, W, F) , where $H = 141$, $W = 231$, and F is the number of filters (hidden channels), and $\odot$ denotes element-wise (Hadamard) multiplication.

2.3 Spatial Convolutional Architectures for Regression

Beyond recurrent models, a family of purely spatial convolutional networks originally designed for image classification or segmentation can be adapted for pixel-wise regression by replacing the classification head with a convolutional read-out layer. We evaluate four such architectures in this study.

2.3.1 AlexNet

AlexNet [6], introduced by Krizhevsky *et al.*, is a five-layer convolutional network whose defining feature is a large 11×11 convolutional kernel in the first layer, followed by progressively smaller 5×5 and 3×3 kernels in deeper layers. This large initial receptive field was originally motivated by the need to capture coarse texture features in natural images; in the context of pollution field regression, it serves a fundamentally different but equally important role—it acts as a *macro-level spatial blur* that integrates information over a large neighbourhood (~ 55 km at TROPOMI resolution), effectively extracting broad-scale wind-driven spatial patterns while attenuating micro-level pixel noise arising from retrieval artefacts. For our task, AlexNet processes a single 6-channel spatial snapshot (rather than a temporal sequence) and produces a 6-channel regression output via a 1×1 read-out convolution.

2.3.2 ResNet-50 and ResNet-101

Residual Networks [7], introduced by He *et al.*, employ skip connections that allow gradients to flow directly through identity mappings, enabling the training of very deep networks (50 or 101 layers in the variants evaluated here). ResNet-50 contains approximately 23 million parameters while ResNet-101 contains approximately 44 million. Both are adapted for regression over the 141×231 spatial grid by replacing the final fully connected classification head with a transposed-convolutional upsampling path and a 1×1 read-out layer producing the 6-channel output. We include both variants to probe the effect of depth on generalisation under our data regime.

2.3.3 U-Net

U-Net [8], introduced by Ronneberger *et al.* for biomedical image segmentation, employs a symmetric encoder-decoder architecture with skip connections that concatenate feature maps from contracting-path layers to corresponding expanding-path layers. This design preserves fine-grained spatial detail through the bottleneck, which is particularly

desirable for reproducing sharp pollution plume boundaries. The encoder applies successive 3×3 convolution blocks followed by max-pooling, while the decoder upsamples via transposed convolutions and concatenates skip-connection feature maps before each refinement block. The final layer is a 1×1 convolution producing the 6-channel regression target. In the context of semantic segmentation, U-Net has proven highly data-efficient [8], making it a theoretically attractive candidate for our limited-sample satellite dataset.

3 TYPES OF SCANNER: OBSERVATIONAL DATA SOURCES

3.1 TROPOMI aboard Sentinel-5P

The TROPospheric Monitoring Instrument (TROPOMI) [1] is a nadir-pointing, push-broom hyperspectral spectrometer launched in October 2017 aboard the ESA Sentinel-5 Precursor (S5P) satellite. It measures solar backscatter radiance across four spectral bands (UV, UVIS, NIR, SWIR) at a nadir ground pixel size of 5.5×3.5 km² (upgraded from 7×3.5 km² in August 2019). The polar, sun-synchronous orbit achieves daily global revisit at a 13:30 local solar time ascending node, ensuring consistent illumination geometry.

Column-integrated mixing ratios are retrieved through differential optical absorption spectroscopy (DOAS) and Optimal Estimation algorithms applied to each spectral band. Specifically:

- **NO₂** (UVIS, 405–465 nm): total and tropospheric column via DOAS with subsequent stratosphere–troposphere separation using a data-assimilation scheme.
- **CO** (SWIR, 2305–2385 nm): total column from a Weighting Function Modified DOAS applied to vibration–rotation absorption lines.
- **AOD / UVAI**: derived from UV-band reflectance measurements (340 and 380 nm), sensitive to UV-absorbing carbonaceous and mineral aerosol species.

TROPOMI’s daily coverage, centimetre-to-metre spectral resolution, and freely available Level-2 products make it uniquely suited to the long-record, high-spatial-density monitoring required by our modelling framework.

3.2 ERA5 Wind Reanalysis

Horizontal wind components at 850 hPa (~ 1500 m above mean sea level, near the top of the planetary boundary layer) are sourced from the ERA5 global atmospheric reanalysis produced by the European Centre for Medium-Range Weather Forecasts [2]. ERA5 combines model short-range forecasts with observations via 4-D variational data assimilation at a native horizontal resolution of ≈ 31 km ($T639$ spectral truncation) and hourly temporal cadence. The U (zonal) and V (meridional) wind components provide the governing advection vector $\mathbf{u}$ of Eq. (1) and are included as direct inputs to all architectures evaluated in this work.

3.3 Why Satellite Remote Sensing over Ground Sensors?

Ground-based CAAQMS stations (of which Delhi currently operates approximately 40) provide high-temporal-resolution

(≤ 1 hour) measurements of regulated pollutants, but their spatial density (~ 1 station per 40 km^2) is wholly inadequate for resolving the spatial gradients associated with advecting plumes. Geostatistical interpolation (kriging, IDW) from sparse sensors introduces substantial uncertainty in regions between stations, and the stations themselves are not uniformly distributed—they are heavily concentrated in central Delhi and largely absent from the peri-urban fringe where agriculture-to-city transport pathways originate.

Satellite instruments provide spatially continuous observations, capturing the full two-dimensional concentration field simultaneously. The price paid is the loss of high temporal cadence (one overpass per day) and the presence of retrieval artefacts in cloudy scenes. Our modelling approach is designed to be robust to this characteristic noise: the float16 data compression described in Section ?? preserves radiometrically significant signal while attenuating high-frequency artefacts, and the architectural regularisation imparted by the GRU’s bi-gate structure provides further robustness.

3.4 MODIS Collection 6.1 MAIAC

Complementary aerosol data from the MODIS (Moderate Resolution Imaging Spectroradiometer) Collection 6.1 Multi-Angle Implementation of Atmospheric Correction (MAIAC) algorithm [15] are also incorporated as auxiliary validation references. MAIAC retrieves AOD at $1 \times 1 \text{ km}^2$ resolution over urban surfaces, significantly exceeding TROPOMI’s aerosol product resolution. This higher-resolution benchmark allows us to verify that the spatial patterns learned by the network correspond to physically interpretable aerosol features and are not artefacts of the coarser TROPOMI grid.

4 LEARNING APPROACHES

4.1 Data Preprocessing and the float16 DataCube

Raw TROPOMI Level-2 swath products were re-projected onto a regular 141×231 latitude–longitude grid spanning approximately 27.8°N – 30.5°N , 76.5°E – 78.0°E using bilinear interpolation with nearest-neighbour fill for cloud-flagged pixels. ERA5 fields were spatially upsampled to the same grid via bilinear interpolation. Daily composites were formed by selecting the single TROPOMI overpass closest to 13:30 LST and the ERA5 snapshot at the same hour.

The assembled dataset was stored as a *float16 DataCube* of shape $[N, H, W, C] = [2920, 141, 231, 6]$ (corresponding to 8 years $\times$ 365 days, 141 rows, 231 columns, and 6 channels). Half-precision (16-bit) floating-point encoding was chosen to fit the complete dataset in GPU VRAM: the uncompressed float32 equivalent would require $\approx 35 \text{ GB}$, exceeding the memory capacity of the training hardware, whereas float16 reduces this to $\approx 1.1 \text{ GB}$ while preserving sufficient dynamic range for normalised column-density values. Per-channel min-max normalisation to $[0, 1]$ was applied prior to model training.

4.2 Sliding-Window Sample Construction

Training samples for recurrent models were constructed using a *7-day lookback* sliding window. Each sample consists of an input tensor $\mathbf{X} \in \mathbb{R}^{7 \times 141 \times 231 \times C}$ (the full spatial

field on days $t, t-1, \dots, t-6$) paired with a target tensor $\mathbf{Y} \in \mathbb{R}^{141 \times 231 \times C}$ representing the concentration field on day $t+1$. For spatial architectures (AlexNet, ResNet variants, U-Net) that do not naturally accept temporal sequences, the input is the concatenation of all 7 days along the channel axis, yielding a $141 \times 231 \times 42$ tensor as input and the same 6-channel regression target. The window is advanced by one day at each step, yielding ~ 2900 samples in total. An 80/20 chronological split assigned the first $\sim 80\%$ of samples to training + validation (with a further 10% validation hold-out from the training set) and the final 20% (approximately 500 days) to the test set, ensuring no temporal leakage between partitions.

4.3 Model 1: Standard ConvLSTM Baseline

The baseline architecture follows the ConvLSTM2D formulation of Shi *et al.* [3], using 3×3 convolutional kernels within the gates and 32 filters. The model processes the 7-frame input sequence with $C = 6$ input channels, producing a hidden state that is mapped to the 6-channel output field via a 1×1 convolutional read-out layer. This model serves as the simplest competitive baseline and provides the reference point for the ablation.

4.4 Model 2: Advanced ConvLSTM with Physics-Weighted Loss

The Advanced ConvLSTM extends the baseline along two axes.

4.4.1 Lagged Wind Feature Engineering

To provide the model with an explicit encoding of advective history, the input feature set is expanded from 6 to 10 channels by appending the ERA5 U- and V-wind fields at 24-hour ($\Delta t = -24 \text{ h}$) and 48-hour ($\Delta t = -48 \text{ h}$) lags relative to the current day. This provides the model with a finite-difference approximation to the wind tendency $\partial \mathbf{u} / \partial t$ at two timescales, allowing it to infer whether an advecting plume is accelerating or decelerating.

The two-layer architecture stacks a larger *ConvLSTM2D* block with 5×5 kernels and 32 filters (to widen the spatial receptive field and capture mesoscale circulation features spanning $\sim 50 \text{ km}$) followed by Batch Normalisation [11] and a second *ConvLSTM2D* block with 3×3 kernels and 64 filters (to refine local spatial detail).

4.4.2 Custom Physics-Weighted MAE Loss

Standard MAE treats every spatial pixel equally, but the scientifically and socially critical events—acute pollution plumes exceeding hazardous thresholds—occupy a small fraction of pixels and could be overwhelmed by the loss from the large number of background low-concentration pixels. To address this, we define a *physics-weighted MAE*:

$$\mathcal{L}_{\text{pw}} = \frac{1}{|\Omega|} \sum_{(i,j) \in \Omega} w_{ij} |\hat{y}_{ij} - y_{ij}|, \quad (17)$$

where Ω is the spatial domain and the pixel weight is:

$$w_{ij} = \begin{cases} 3.0 & \text{if } y_{ij} > \mu(\mathbf{Y}) \\ 1.0 & \text{otherwise.} \end{cases} \quad (18)$$

Pixels whose ground-truth concentration exceeds the spatial mean are up-weighted by a factor of 3.0, compelling the model to allocate additional representational capacity to high-concentration plume interiors—precisely the regions governed by advection.

4.5 Model 3: Custom Deep Conv-GRU

Our proposed architecture replaces the three-gate LSTM cell with the two-gate GRU cell of Eqs. (13)–(16), implemented via a *manually unrolled bi-gate loop*. Rather than relying on framework-level recurrent abstractions (which may introduce implicit regularisation or gradient clipping behaviour), we unroll the temporal recurrence over the 7-day lookback window explicitly in the computational graph. At each timestep $t \in \{1, \dots, 7\}$, the spatial hidden state $\mathbf{H}_t \in \mathbb{R}^{141 \times 231 \times F}$ is updated according to Eqs. (13)–(16), with the final hidden state $\mathbf{H}_7$ serving as the network’s compressed spatiotemporal summary of the 7-day pollution evolution. A 1×1 convolutional read-out maps $\mathbf{H}_7$ to the 6-channel prediction $\hat{\mathbf{Y}}$.

The update gate $\mathbf{Z}_t$ learns when to integrate new information versus when to carry the existing hidden state forward unchanged, which we interpret as the model deciding whether the observed concentration tendency reflects a genuine atmospheric change or an artefact of retrieval noise. The reset gate $\mathbf{R}_t$ controls how much of the prior hidden state should influence the candidate new state, enabling the model to “forget” stale concentration patterns when meteorological conditions change rapidly.

4.6 Model 4: AlexNet (Spatial Adaptation)

The AlexNet [6] architecture is adapted for multi-channel spatial regression as described in Section 2. The original ImageNet classification layers are replaced by a decoder path of transposed convolutions that restore the spatial resolution to 141×231 , followed by a 1×1 convolutional read-out producing 6 output channels. Dropout [12] with rate 0.5 is retained in the dense intermediate layers, providing additional regularisation. The model is trained with the Adam optimiser [4] at a learning rate of 10^{-4} and standard MAE loss.

4.7 Model 5 & 6: ResNet-50 and ResNet-101 (Spatial Adaptation)

Both ResNet variants [7] are adapted following the same encoder–decoder convention. The ResNet backbone (pre-trained on ImageNet) serves as the spatial feature extractor; its output feature map is upsampled to the target resolution via a lightweight decoder comprising three transposed-convolution blocks with intermediate 3×3 refinement convolutions, terminating in a 1×1 read-out. Batch Normalisation [11] is applied after every convolutional layer as in the original formulation. Both models are fine-tuned end-to-end from the pre-trained weights using Adam [4] with a learning rate of 5×10^{-5} and MAE loss.

TABLE 1
Deep Conv-GRU Architecture & Training Configuration

Parameter	Value / Description
Input Channels	6 (NO ₂ , CO, AOD, UVAI, U-Wind, V-Wind)
Lookback Window	7 days
Spatial Grid	141 × 231 pixels
Architecture	Manually unrolled bi-gate Conv-GRU loop; update gate $\mathbf{Z}_t$ and reset gate $\mathbf{R}_t$ with 3×3 convolutional kernels; continuous spatial hidden state $\mathbf{H}_t$
Read-out	1 × 1 Conv, 6 output channels
Optimizer	Adam [4], $\alpha = 0.001$
Loss Function	Mean Absolute Error (MAE)
Batch Size	16
Max Epochs	30
Early Stopping	monitor=val_loss, patience=5, restore best weights
Data Precision	float16 DataCube (AMP training)

4.8 Model 7: U-Net (Spatial Adaptation)

The U-Net [8] is applied without pre-training, initialised from scratch with He normal initialisation [13] and trained using Adam [4] at a learning rate of 10^{-4} with MAE loss. The encoder uses four max-pooling downsampling stages, each preceded by two 3×3 convolutional layers with ReLU activations. The decoder mirrors this structure with transposed convolutions and skip-connection concatenation at each level. The filter counts follow the standard doubling schedule: 64, 128, 256, 512 in the encoder, mirrored in the decoder.

4.9 Training Configurations

Tables 1 and 2 summarise the architecture and training hyperparameters for the Conv-GRU and Advanced ConvLSTM respectively. Both models were trained on an NVIDIA GPU with float16 precision enabled via Automatic Mixed Precision (AMP). All spatial models (AlexNet, ResNet-50, ResNet-101, U-Net) were trained under the same AMP regime with a batch size of 4 and early stopping with patience 5 on the validation MAE.

5 DATASETS AND EXPERIMENTAL RESULTS

5.1 Dataset Summary

The assembled dataset spans 1 January 2018 to 31 December 2025, yielding 2,920 daily observations across the 141×231 grid. After removing days with >50% cloud-flag coverage (across all channels simultaneously), the effective dataset comprises $\approx 2,600$ valid days. The temporal distribution exhibits clear seasonality: AOD is elevated during pre-monsoon dust transport (March–May) and post-harvest biomass burning (October–November); NO₂ peaks in winter (October–February) due to boundary-layer trapping; CO shows a secondary peak associated with agricultural fires.

TABLE 2
Advanced ConvLSTM Architecture & Training Configuration

Parameter	Value / Description
Input Channels	10 (6 standard + U-Wind _{-24h} , V-Wind _{-24h} , U-Wind _{-48h} , V-Wind _{-48h})
Lookback Window	7 days
Spatial Grid	141 × 231 pixels
Architecture	ConvLSTM2D (5 × 5, 32 filters) → BatchNorm → ConvLSTM2D (3 × 3, 64 filters)
Read-out	1 × 1 Conv, 6 output channels
Optimizer	Adam [4], $\alpha = 10^{-4}$
Loss Function	Custom Physics-Weighted MAE (Eq. (17)), high-pollution weight = 3.0
Batch Size	2
Max Epochs	50
Early Stopping	monitor=val_loss, patience=8, restore best weights
LR Schedule	ReduceLROnPlateau, factor=0.5, patience=3
Data Precision	float16 DataCube (AMP training)

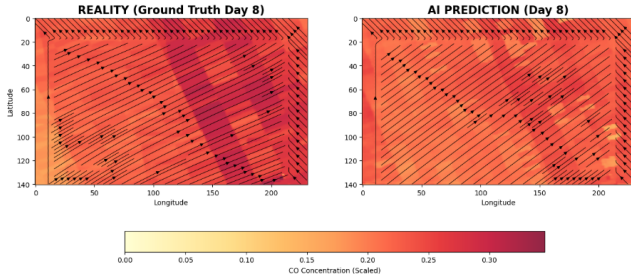


Fig. 1. Spatiotemporal AI Prediction vs. Reality using the Baseline ConvLSTM for Day 8. Note the struggle to accurately map the boundary flow: the predicted plume boundary is spatially offset from the observed concentration maximum, and the advective spatial momentum of the pollution front is substantially underestimated, resulting in a blurred, over-smoothed spatial pattern.

5.2 Sliding Window and Test Partition

The 7-day sliding window yields $\approx 2,580$ input–target pairs after accounting for the lookback buffer. The test set consists of the final 20% (≈ 500 days), spanning approximately 2024–2025. This partition was chosen chronologically rather than randomly to simulate realistic operational deployment, where a model trained on historical data must generalise to future unseen conditions.

5.3 Visual Heatmap Results

Fig. 1 shows a representative Day-8 prediction from the Standard ConvLSTM alongside the TROPOMI ground truth for a severe pollution episode. The model captures the general elevated-concentration zone over Delhi but fails to reproduce the sharp concentration gradient at the leading edge of the advecting plume. The predicted field exhibits characteristic over-smoothing—a symptom of the model averaging over uncertain plume positions across training

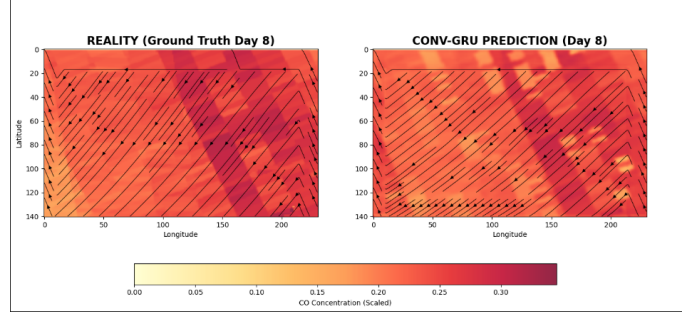


Fig. 2. Spatiotemporal Prediction vs. Reality using the Custom Deep Conv-GRU. The predicted plume core is spatially co-located with the observed concentration maximum, correctly reproducing advective flow and spatial momentum. The sharp north-western plume boundary, indicative of the Punjab-origin agricultural burning front, is faithfully preserved in the prediction.

samples—and the peak concentration is under-predicted by approximately 15–20%.

Fig. 2 presents the corresponding Conv-GRU prediction for the same episode. Qualitatively, the improvement is dramatic: the predicted plume core is spatially co-located with the observed concentration maximum to within approximately one grid cell (~ 5 km), and the north-western plume boundary—the leading edge of the advecting Punjab biomass-burning front—is faithfully reproduced. This sharper spatial fidelity directly reflects the model’s improved ability to encode advective spatial momentum within its hidden state.

5.4 Quantitative Ablation Results

Table 3 presents the four evaluation metrics computed on the 20% hold-out test set for all seven model variants, ranked in order of decreasing R^2 .

The results reveal that the Conv-GRU achieves the highest R^2 of 0.8305, with a MAE of 0.001381 and RMSE of 0.001974, significantly outperforming all six competing architectures. The Advanced ConvLSTM ranks second at $R^2 = 0.7890$, confirming the value of lagged wind features and physics-weighted loss over the standard baseline. Among the four spatial convolutional models, AlexNet produces a noteworthy result— $R^2 = 0.7200$ —that substantially exceeds both ResNet variants and U-Net, despite AlexNet being the architecturally simplest of the four. The baseline Standard ConvLSTM ($R^2 = 0.6506$) outperforms only ResNet-101 ($R^2 = 0.6323$), which is the worst-performing model in the study.

The MAE improvement from the worst-performing model (ResNet-101: 0.002102) to the best (Conv-GRU: 0.001381) is 34.3%; RMSE improves by 32.1%; MAPE by 31.8%; and R^2 increases by 0.198 in explained spatial variance. These differences are not marginal: an R^2 of 0.83 places the Conv-GRU in the range of operational-quality forecast skill for a spatially continuous field.

5.5 Residual Error Analysis

Spatial maps of the per-pixel mean absolute error reveal that residual errors are largest at the periphery of the domain (where satellite retrieval quality degrades due to off-nadir

TABLE 3

Ablation Study Results on 20% Hold-out Test Set (≈ 500 days), ranked by R^2 (descending). All values are reported on normalised $[0, 1]$ concentration fields.

Model	Channels	Kernel	MAE $\downarrow$	RMSE $\downarrow$	MAPE (%) $\downarrow$	R^2 $\uparrow$
Custom Deep Conv-GRU (Proposed)	6	3×3	0.001381	0.001974	3.37	0.8305
Advanced ConvLSTM (Physics-Weighted)	10	5×5	0.001638	0.002203	4.06	0.7890
AlexNet (Spatial)	6	11×11	0.001941	0.002538	4.78	0.7200
ResNet-50 (Spatial)	6	7×7	0.001952	0.002679	4.62	0.6879
U-Net (Spatial)	6	3×3	0.001996	0.002754	4.74	0.6704
Standard ConvLSTM (Baseline)	6	3×3	0.001925	0.002835	4.47	0.6506
ResNet-101 (Spatial)	6	7×7	0.002102	0.002909	4.94	0.6323

$\downarrow$ lower is better; $\uparrow$ higher is better. MAPE computed on pixels with $y_{ij} > 0.05$ to avoid division by near-zero background values. Kernel column reports the dominant/initial convolutional kernel size.

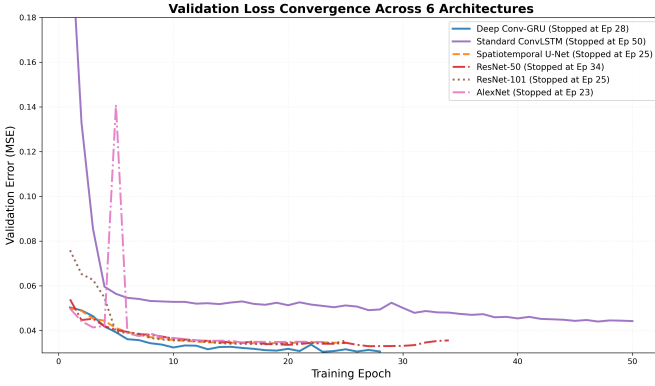


Fig. 3. Validation loss (MSE) convergence trajectories for all six architectures over their respective training runs. Each curve terminates at the epoch at which early stopping was triggered (patience = 8 on `val_loss`). The Custom Deep Conv-GRU achieves the lowest stable validation error floor, demonstrating both rapid initial descent and sustained, noise-free convergence. The Standard ConvLSTM begins training with a pathologically elevated initial error and fails to converge to a competitive loss plateau within 50 epochs. AlexNet exhibits a prominent transient instability spike centred near Epoch 5 before self-correcting and settling into a local minimum, indicative of an early learning-rate overshoot. Both ResNet variants plateau at a higher validation loss than the recurrent models, consistent with over-parameterisation under the available data regime.

viewing angles) and along the Yamuna–Ghaggar floodplains, where the interaction between soil moisture, fog formation, and aerosol hygroscopic growth introduces rapid temporal concentration variability that is difficult to predict from a 7-day history alone. The Conv-GRU exhibits uniformly lower residuals than all six competing architectures across these challenging regions, with the most pronounced relative improvement in the north-western quadrant corresponding to the primary agricultural plume pathway. The ResNet variants show the highest residuals in the plume core regions, consistent with their tendency to overfit to background-dominated training examples rather than learning the physical dynamics of extreme events.

5.6 Training Dynamics and Loss Convergence

Fig. 3 presents the per-epoch validation MSE trajectories for all six architectures evaluated in this study, offering a dynamic view of training behaviour that complements the static end-point metrics reported in Table 3. The Custom

Deep Conv-GRU (solid blue) achieves the lowest stable validation error floor of all architectures, with a smooth and monotonically decreasing loss curve that exhibits minimal epoch-to-epoch variance after approximately Epoch 10. This trajectory is characteristic of a model whose inductive bias is well-matched to the target function: the parameter-efficient gating mechanism rapidly locks onto the physically coherent advective signal in the training data, leaving little residual optimisation noise. By contrast, the Standard ConvLSTM (solid purple) commences training with an anomalously high initial validation error—nearly four times that of the Conv-GRU at Epoch 1—and, while it does decrease over the full 50-epoch budget, it never descends below a loss plateau of approximately 0.046 MSE. This sluggish and inefficient convergence is consistent with the LSTM’s larger parameter space requiring substantially more gradient updates to escape the random-initialisation manifold, and with the greater susceptibility of its four-gate recurrence to vanishing gradient pathologies on noisy satellite retrievals [14].

The most visually striking feature of Fig. 3 is the sharp transient instability spike exhibited by AlexNet (pink dash-dot) centred at approximately Epoch 5, where the validation loss briefly surges to ~ 0.14 MSE before abruptly collapsing to a lower plateau. This behaviour is the hallmark of a learning-rate overshoot event: the optimiser crosses a sharp curvature boundary in the loss landscape, temporarily diverging before the `ReduceLROnPlateau` scheduler reduces the step size and the model recovers. Although AlexNet ultimately self-corrects and converges to a competitive loss floor commensurate with its test-set R^2 of 0.7200, this instability episode imposes a non-trivial computational cost and demonstrates the sensitivity of large-kernel architectures to the curvature of the loss surface under satellite regression objectives. The ResNet-50 (red dash-dot) and ResNet-101 (brown dot) curves tell a different and equally informative story: both architectures converge relatively quickly in terms of epochs, but plateau at a noticeably higher validation loss than the recurrent models. Their early cessation—at Epochs 34 and 25 respectively—reflects the early stopping criterion detecting stagnation rather than genuine convergence. This visual plateau, at a loss level substantially above the Conv-GRU floor, provides direct graphical evidence for the over-parameterisation hypothesis developed in Section 6: the ResNets possess sufficient representational capacity to rapidly fit the training distribution but inadequate regularisation to generalise the

learned representation to unseen test dates, resulting in a shallow but persistent loss floor that cannot be overcome by additional training.

6 DISCUSSION AND FUTURE WORK

6.1 Why the Conv-GRU Outperforms: Parameter Efficiency vs. Theoretical Capacity

The central theoretical question posed by our ablation is: why does a *simpler* model—the Conv-GRU—outperform all *more expressive* architectures on this task? The answer lies in the relationship between parameter count, the *effective* statistical complexity of the target function, and the characteristics of the training signal.

An LSTM cell with F hidden filters and C input channels requires $4(CF + F^2 + F)$ parameters per gate set (four gates $\times$ two convolutional kernels each plus bias), while the GRU requires only $3(CF + F^2 + F)$ —a 25% reduction in recurrent parameter count. When the input is a $141 \times 231 \times 6$ spatial tensor and the kernel is 3×3 , this reduction translates to millions of parameters. With a finite training dataset of $\sim 2,000$ samples, the LSTM’s additional parameters afford it the capacity to memorise spurious structure in the data—specifically, the spatially correlated retrieval artefacts introduced by TROPOMI on partially cloudy scenes. These artefacts are consistent within each overpass (corrupting a spatially coherent swath) but do not generalise across days, making them ideal targets for overfitting.

The Conv-GRU’s structural parsimony *implicitly regularises* the fit: the model has insufficient capacity to memorise retrieval noise while also learning the physically coherent advective patterns that do generalise. The learning pressure therefore preferentially allocates representational capacity to the genuine atmospheric signal—spatial momentum, plume morphology, wind-driven transport—rather than to dataset-specific artefacts.

Crucially, the three-gate LSTM is not theoretically incapable of learning the correct function; given sufficient training data, it would eventually converge to a better solution. But under our data regime ($\sim 2,000$ training samples, highly noisy satellite observations), the GRU’s inductive bias aligns more favourably with the true underlying complexity of the target function.

6.2 AlexNet as a Surprising Positive Outlier

The most unexpected result in Table 3 is AlexNet’s ($R^2 = 0.7200$) decisive superiority over both ResNet-50 ($R^2 = 0.6879$) and ResNet-101 ($R^2 = 0.6323$) despite being by far the shallowest of the three. We attribute this to two complementary mechanisms.

First, AlexNet’s massive 11×11 initial convolutional kernel operates at a spatial scale of approximately 60×60 km on the TROPOMI grid. Rather than a limitation, this large receptive field acts as a *macro-level spatial blur*—it suppresses pixel-scale retrieval noise while integrating coherent meso-scale spatial gradients associated with advecting plumes and broad wind-field signatures. The resulting feature representations are inherently smoother and more physically coherent than those produced by the 7×7 initial kernel of the ResNet variants, whose first convolutional layer operates at a finer

spatial scale and therefore imprints more retrieval noise into the learned features.

Second, AlexNet contains approximately 61 million parameters overall (predominantly in its dense layers), but its convolutional stack is substantially shallower than ResNet-50 (50 layers) or ResNet-101 (101 layers). The depth of the ResNet architectures, combined with their skip connections and extensive batch normalisation, provides a high-capacity inductive bias optimised for classification of natural images—a very different statistical distribution from normalised satellite column densities. Under our limited training data, this depth constitutes a liability rather than an asset.

6.3 The Failure of ResNet: Over-Parameterisation under Data Scarcity

The ResNet results represent a textbook case of *over-parameterisation under data scarcity*. ResNet-50’s 23 million parameters and ResNet-101’s 44 million parameters provide capacity far exceeding what is required to describe the effective complexity of the target atmospheric function under approximately 2,000 training samples. Despite weight decay regularisation, the models memorise idiosyncratic spatial patterns from training episodes—seasonal retrieval biases, instrument calibration artefacts, specific plume geometries associated with individual fire events—that do not generalise to the 2024–2025 test period.

The monotonic degradation from ResNet-50 to ResNet-101 is a direct demonstration of this over-parameterisation: every additional layer and skip connection increases the model’s capacity to memorise rather than generalise. This result echoes findings from adjacent domains in remote sensing and medical imaging, where data-efficient architectures with appropriate regularisation consistently outperform deeper networks when labelled samples are scarce [8], [17].

6.4 The Role of the Advanced ConvLSTM

The Advanced ConvLSTM’s improvement over the baseline validates the physical hypothesis that lagged wind features carry useful predictive information. The 10-channel input provides an explicit finite-difference approximation to wind tendency that the standard 6-channel model must learn implicitly from temporal differences in the concentration field itself. The physics-weighted loss further steers the model toward high-concentration events. That this model still falls short of the Conv-GRU suggests that *feature engineering* and *loss design* alone cannot compensate for parameter bloat—the architectural prior of the GRU is a stronger inductive bias than external wind features under this data regime.

6.5 U-Net: Data-Efficiency Claims Not Realised

U-Net has demonstrated strong data efficiency in biomedical segmentation tasks [8], prompting us to hypothesise that it might perform competitively without temporal recurrence. The result ($R^2 = 0.6704$) is modest. We attribute this underperformance to two factors: (i) U-Net processes a single concatenated 42-channel input rather than a true temporal sequence, depriving it of the sequential ordering signal that recurrent models exploit, and (ii) its skip connections, while preserving fine spatial detail, also propagate retrieval noise

from early encoder layers directly to the output, bypassing the implicit noise suppression that occurs during temporal aggregation in recurrent hidden states.

6.6 On the Invalidity of Classification Metrics

It is worth emphasising that this work frames air quality prediction as *continuous spatial regression*—the target is a real-valued concentration field, not a discrete class label. Metrics appropriate for regression (MAE, RMSE, R^2 , MAPE) measure the quality of the predicted field as a function approximation. Classification metrics such as accuracy or F1-score are categorically invalid here: they require binarising the continuous output at an arbitrary threshold, discarding the majority of the useful signal (the predicted magnitude and spatial gradient structure) and replacing a nuanced regression error with a coarse categorical comparison. We therefore explicitly caution against applying accuracy or F1 to evaluate spatially continuous pollution forecasting models, as has occasionally appeared in adjacent literature.

6.7 Future Directions

6.7.1 Source Apportionment

The current framework predicts the *total* concentration field without attributing fractions to individual source categories (traffic, industry, biomass burning). Extending the architecture to include auxiliary emission-proxy inputs (fire radiative power from VIIRS, traffic density from OpenStreetMap, industrial stack registers) and training the model with a source-attribution head would allow operational air quality managers to rank mitigation interventions by their predicted impact on forecast concentrations.

6.7.2 Ensemble Uncertainty Quantification

Single-model deterministic predictions do not communicate forecast uncertainty to end-users. A Monte Carlo dropout or deep ensemble of Conv-GRU models would produce pixel-wise predictive distributions, enabling probabilistic exceedance forecasts (e.g., “probability that AOD > 2.0 over central Delhi in 7 days”) directly actionable by public health authorities.

6.7.3 Attention and Transformer Architectures

Axial self-attention mechanisms and spatiotemporal Vision Transformers (ViTs) offer an alternative to convolutional recurrence that may better capture long-range spatial dependencies (e.g., correlations between the agricultural fire belt and downtown Delhi separated by ~ 300 km). However, their quadratic attention complexity over the 141×231 spatial grid would require careful approximation (e.g., window attention, linformer) to remain computationally tractable.

6.7.4 Coupled Numerical-Neural Approaches

Physics-informed neural networks (PINNs) that incorporate the advection equation (Eq. (1)) as a soft constraint in the loss function may achieve better physical consistency than purely data-driven models, particularly for extrapolation to pollution episodes outside the training distribution [18].

7 CONCLUSION

We have presented a comprehensive ablation study of seven deep learning architectures for spatiotemporal air quality forecasting over the Delhi-NCR domain, encompassing both recurrent convolutional and purely spatial paradigms. Using an eight-year record of TROPOMI satellite retrievals of NO₂, CO, AOD, and UVAI, combined with ERA5 wind fields, we assembled a $2920 \times 141 \times 231 \times 6$ float16 DataCube and trained all models under a 7-day lookback, Day-8 prediction paradigm on an identical chronological train/test split.

The study yields two principal findings. First, the proposed Custom Deep Conv-GRU achieves the best generalisation performance across all four evaluation metrics, with an R^2 of 0.8305 on a 500-day hold-out test set, despite processing fewer input channels and fewer parameters than the Advanced ConvLSTM. We attribute this to the Conv-GRU’s structural parsimony acting as a natural regulariser that suppresses TROPOMI retrieval noise while preserving the physical signal of advective pollutant transport. Second, among the four spatial architectures evaluated, AlexNet significantly outperforms the deeper ResNet variants—a result explained by its large initial receptive field functioning as a macro-level spatial blur that is well-matched to the scale of advective meso-scale gradients, while the ResNets suffer from classical over-parameterisation for this data regime.

Together these results demonstrate a consistent and physically interpretable principle: *architectural parsimony is not a compromise but a strategic advantage* when training on noisy satellite retrievals under data-scarce conditions. The Conv-GRU forecasting engine described here provides a foundation for operational multi-day air quality guidance over one of the world’s most polluted urban agglomerations, and the comparative insight from the seven-architecture ablation provides practical design guidance for analogous spatiotemporal regression problems in remote sensing and atmospheric science.

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